

Submission C solution to Problem 4

Abstract

We prove the stated estimate as an immediate consequence of Guth's product-domain form of his area-contraction estimates for rectangles. All constants below are positive dimensional constants; in this problem the dimension is fixed equal to 4, so the final constant is universal.

1 A product estimate of Guth

We shall use the following special case of Guth's product-domain estimate. It is the case $n = 4$, $k = 2$, $\dim X = 1$, and $\dim Y = 2$ of Proposition 9.1 in Guth's paper [1, Proposition 9.1]. The proof in [1] is obtained from the isoperimetric profile of rectangles, the Almgren–Gromov complex-of-cycles formalism, and the tightening argument developed in Sections 1, 3, 4, and 6 of that paper; a fuller account appears in Guth's thesis [2].

Quoted theorem 1 (Guth, specialized product estimate). *There are constants $\varepsilon_0 > 0$ and $c_0 > 0$ with the following property.*

Let

$$X = [0, a], \quad I = [0, b], \quad Y = [0, u] \times [0, v],$$

with the product Euclidean metric, and let $S \subset \mathbb{R}^4$ be a rectangle with side lengths

$$S_1 \leq S_2 \leq S_3 \leq S_4.$$

Suppose

$$F : (X \times I \times Y, \partial(X \times I \times Y)) \longrightarrow (S, \partial S)$$

is piecewise smooth, has non-zero degree, and satisfies

$$\mathrm{Dil}_2(F) \leq 1.$$

If

$$a \leq \varepsilon_0 S_1 \quad \text{and} \quad uv \leq \varepsilon_0 S_3 S_4,$$

then

$$\mathrm{Vol}_4(X \times I \times Y) = abuv \geq c_0 S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Remark 1. *Guth states his estimates for Lipschitz/PL complexes of cycles. A piecewise smooth map on a compact piecewise smooth rectangle is admissible in that framework, since push-forwards of integral Lipschitz chains are defined simplexwise. Alternatively, one may approximate a piecewise smooth relative map by smooth relative maps with 2-dilation increased by an arbitrarily small factor and then let the approximation parameter tend to zero. The relative degree is unchanged under such approximation.*

2 Proof of the required estimate

Theorem 1. *There is a universal constant $\kappa > 0$ such that the following holds. Let $R \subset \mathbb{R}^4$ and $S \subset \mathbb{R}^4$ be rectangles with side lengths*

$$R_1 \leq R_2 \leq R_3 \leq R_4, \quad S_1 \leq S_2 \leq S_3 \leq S_4.$$

Suppose

$$f : (R, \partial R) \longrightarrow (S, \partial S)$$

is piecewise smooth, has degree 1, and satisfies $\text{Dil}_2(f) \leq 1$. If

$$R_1 \leq \kappa S_1,$$

then either

$$R_3 R_4 > \kappa S_3 S_4,$$

or

$$R_1 R_2 R_3 R_4 > \kappa S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Proof. Let ε_0, c_0 be the constants from Theorem 1. Define

$$\kappa := \frac{1}{2} \min\{\varepsilon_0, c_0\}.$$

Assume

$$R_1 \leq \kappa S_1.$$

If

$$R_3 R_4 > \kappa S_3 S_4,$$

then the first desired alternative holds, and there is nothing to prove.

Thus assume instead that

$$R_3 R_4 \leq \kappa S_3 S_4.$$

Write the domain rectangle as the product

$$R = [0, R_1] \times [0, R_2] \times ([0, R_3] \times [0, R_4]).$$

Set

$$X = [0, R_1], \quad I = [0, R_2], \quad Y = [0, R_3] \times [0, R_4].$$

Then

$$\text{Vol}_1(X) = R_1 \leq \kappa S_1 \leq \varepsilon_0 S_1$$

and

$$\text{Vol}_2(Y) = R_3 R_4 \leq \kappa S_3 S_4 \leq \varepsilon_0 S_3 S_4.$$

The map f is a degree-one map of pairs

$$f : (X \times I \times Y, \partial(X \times I \times Y)) \longrightarrow (S, \partial S)$$

with $\text{Dil}_2(f) \leq 1$. Hence Theorem 1 applies and gives

$$R_1 R_2 R_3 R_4 = \text{Vol}_4(X \times I \times Y) \geq c_0 S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Since $\kappa \leq c_0/2$, this implies

$$R_1 R_2 R_3 R_4 > \kappa S_1 S_2^{1/2} S_3^{3/2} S_4.$$

Therefore the second desired alternative holds. This proves the theorem. $\square$

References

- [1] L. Guth, *Area-expanding embeddings of rectangles*, arXiv:0710.0403, 2007.
- [2] L. Guth, *Area-contracting maps between rectangles*, Ph.D. thesis, Massachusetts Institute of Technology, 2005.
- [3] H. Federer and W. H. Fleming, *Normal and integral currents*, Ann. of Math. (2) **72** (1960), 458–520.
- [4] F. J. Almgren, *The homotopy groups of the integral cycle groups*, Topology **1** (1962), 257–299.
- [5] M. Gromov, *Filling Riemannian manifolds*, J. Differential Geom. **18** (1983), no. 1, 1–147.